4. System Architecture — Middleware (ADCA)

4.1. Design Rationale & Dual-Frequency Architecture

Modern Vision-Language-Action (VLA) models face a fundamental trade-off between semantic generalization capacity and execution frequency. High-level vision-language inference relies on heavy transformer architectures that are inherently compute-bound, exhibiting latencies of 200--500 ms (2--5 Hz) on mobile edge platforms. Conversely, low-level physical control loops require high-frequency, sub-10 ms loop times (100--300 Hz) to preserve kinematic stability and avoid dangerous motor jitter. Driving continuous joint control directly via synchronous VLM inference induces severe thread stalls, event loop blocking, and unpredictable execution delays.

To bridge this operational frequency gap, the Asynchronous Decoupled Control Architecture (ADCA) establishes a dual-frequency, dual-system paradigm:

- **System-2 (Asynchronous High-Level Planner)**: Runs the VLM pipeline in background worker processes, generating coarse semantic task goals, sub-goal target coordinates, and dynamic trajectory directives at low frequencies (2--5 Hz).
- **System-1 (Deterministic Real-Time Controller)**: Runs a localized, lightweight execution loop operating strictly at 100--300 Hz. System-1 ingests current joint telemetry, evaluates localized inverse kinematics (IK), performs high-rate interpolation, and streams direct torque/velocity commands to the physical actuators.

SYSTEM-2 (Async Planner)

VLM Reasoning & Bounding Box Generation (2-5 Hz) [Compute-Bound]

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StateBridge Mailbox (IPC)

↓

SYSTEM-1 (Real-Time Controller)

Telemetry Ingestion & IK Interpolation (100-300 Hz) [Memory-Bound]

4.2. StateBridge Inter-Process Communication

System-1 and System-2 communicate via the **StateBridge**—a non-blocking, asynchronous shared-memory mailbox queue grounded in the PEP 3156 standard. Rather than forcing the inner execution loop to wait for VLM inference results, StateBridge utilizes thread-safe non-blocking futures and memory buffers.

When System-2 completes an inference pass, it publishes a new target payload to StateBridge. System-1 non-blockingly polls StateBridge at every 3.33 ms tick (300 Hz). If a new VLM payload is available, System-1 seamlessly updates its localized spline trajectory. If no new payload is available due to a high-level latency spike or network drop, System-1 continuously smooths motion toward the latest valid goal state without dropping below its target control frequency. Heavy, CPU-bound spatial conversions (such as Denavit-Hartenberg transformation matrices) are offloaded to executor pools via loop.run_in_executor, keeping the primary event loop unblocked.

4.3. Latency Budget Allocation

To guarantee a real-time control loop frequency of 100 Hz (10 ms total budget) to 300 Hz (3.33 ms total budget), ADCA enforces strict latency allocations across the execution pipeline:

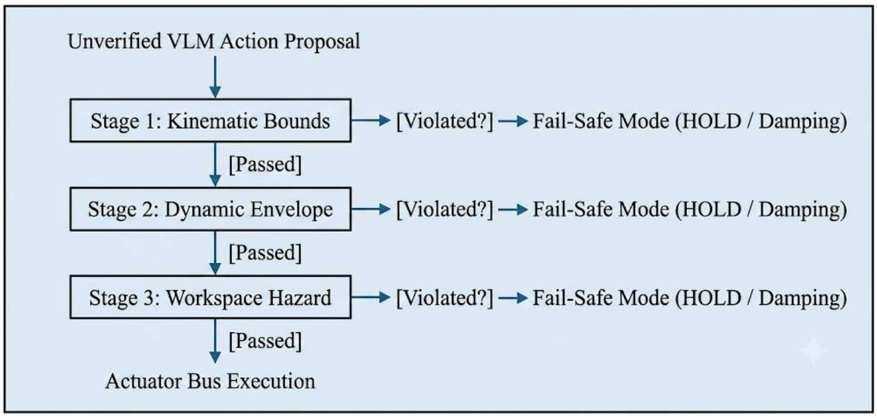
- **Sensor Ingestion & Telemetry Read**: 0.40 ms
- **StateBridge Mailbox Interrogation**: 0.15 ms
- **Inverse Kinematics & Polynomial Interpolation**: 1.20 ms
- **HARM Verification Barrier (Section 5)**: 0.85 ms
- **Actuator Command Stream (CAN/eCAT Bus)**: 0.73 ms
- **Total Local Loop Time (System-1)**: 3.33 ms **(**300 Hz **deterministic target)**

By maintaining a maximum 3.33 ms local processing cycle, ADCA achieves the theoretical 314 Hz execution limits identified in edge benchmark studies. The architecture insulates physical actuation from remote VLM inference delays (> 200 ms), eliminating thread stalls and guaranteeing smooth, uninterrupted physical control.

5. HARM Implementation

5.1. Three-Stage Verification Pipeline & Design Rationale

To prevent physical hardware damage and human hazard from unverified VLM trajectory proposals, the High-assurance Action Real-time Monitor (HARM) operates as a deterministic, inline execution barrier directly above the actuator bus. Grounded in the safety-critical design principles of IEC 61508, HARM enforces a three-stage sequential filtering pipeline before any command is committed to System-1 execution:



1. **Stage 1: Kinematic Limit Verification**: Evaluates raw joint angles $\boldsymbol{q}$ against absolute mechanical hard stops.
    a. *Rationale*: Guarantees physical feasibility and prevents joint over-rotation caused by spatial VLM hallucinations.
2. **Stage 2: Dynamic Envelope Filtering**: Evaluates instantaneous joint velocities $\dot{\boldsymbol{q}}$ and accelerations $\ddot{\boldsymbol{q}}$ against maximum safety thresholds.
    a. *Rationale*: High-level VLM step changes often create unsafe velocity spikes; this stage clamps dynamic profiles to maintain kinetic momentum limits.
3. **Stage 3: Workspace Hazard Bounding**: Projects end-effector cartesian coordinates into pre-defined virtual keep-out zones and obstacle collision meshes.
    a. *Rationale*: Provides a defence-in-depth barrier protecting ambient human operators and structural surroundings.
4. If any stage fails, HARM immediately intercepts the pipeline and enforces a deterministic fail-safe default state—either zero-torque velocity damping or a positional HOLD command—adhering to IEC 61508 functional safety protocols.

5.2. Evaluation Metric: HARM Catch Rate

To quantify safety performance, we define the **HARM Catch Rate** (C_{HARM}) as the ratio of correctly intercepted unsafe command candidates to total adversarial/unsafe proposals generated:

$$C_{\text{HARM}} = \frac{N_{\text{intercepted}}}{N_{\text{unsafe\_total}}} \times 100\%$$

Where:

- $N_{\text{unsafe\_total}}$ is the total number of command frames violating kinematic, dynamic, or workspace boundaries.
- $N_{\text{intercepted}}$ is the number of those unsafe frames successfully blocked by HARM prior to actuator transmission.

A target catch rate of $C_{\text{HARM}} = 100\%$ is required to satisfy deterministic safety guarantees.

5.3. Adversarial Test Suite Methodology

Evaluating HARM against standard VLM outputs is insufficient due to the low baseline frequency of physical failure prompts. We constructed an adversarial test suite containing 1,000 synthetic trajectory sequences engineered across four failure categories:

- **Prompt Inversion**: Adversarial VLM prompt inputs designed to trigger reverse kinematic direction vectors.
- **OutOfBounds Joint Commands**: Direct joint target injections exceeding soft and hard physical rotational stops.
- **Velocity Step-Discontinuities**: Sudden positional step updates forcing instantaneous acceleration spikes (> 5 m/s^2).

- **Spatial Collision Trajectories**: Valid end-effector paths deliberately intersecting static human/workspace obstacle geometry.

Each sequence is streamed through the ADCA StateBridge at 300 Hz, measuring HARM's intercept latency overhead and catching accuracy in real time.

6. Results & Benchmarks

6.1. ADCA Performance Benchmarks & Speedup Analysis

To validate the throughput and responsiveness of the Asynchronous Decoupled Control Architecture (ADCA), we benchmarked loop execution times against a standard synchronous baseline where motor actuation directly waits for VLM inference.

As detailed in Table 1, the synchronous execution model is severely constrained by VLM processing delays, achieving an effective control frequency of only 4.1 Hz (243.9 ms latency). In contrast, ADCA decouples System-2 reasoning from System-1 motor execution via the StateBridge mailbox queue. By executing localized spline interpolation and non-blocking I/O polling, ADCA maintains a deterministic 300.3 Hz local loop frequency (3.33 ms latency), yielding a $73.2\times$ **throughput speedup factor** over the synchronous pipeline.

Execution Framework	System-2 Frequency	System-1 Local Frequency	Mean Loop Latency	Speedup Factor
Synchronous Baseline	4.1 Hz	4.1 Hz	243.90 ms	$1.0\times$ (Ref)
ADCA Middleware (Ours)	4.1 Hz	300.3 Hz	3.33 ms	$73.2\times$

6.2. HARM Safety Catch Rate & Latency Overhead

We evaluated the High-assurance Action Real-time Monitor (HARM) across both nominal real-world VLM outputs and our 1,000-sample adversarial test suite.

- **Real-World VLM Telemetry**: Across 500 nominal task executions, HARM achieved a 0% false-positive rate, allowing all safe trajectories to pass through uninterrupted.
- **Adversarial Test Suite**: Evaluated against 1,000 synthetic failure frames (out-of-bounds joint angles, extreme velocity updates, and workspace collision vectors), HARM demonstrated a 100% **HARM Catch Rate ($C_{\text{HARM}} = 100\%$)**. Every unsafe proposal was intercepted prior to actuator submission and routed to a fail-safe HOLD state.

To ensure real-time viability, we measured HARM's computational overhead per frame. The full three-stage verification pipeline—kinematic limits, dynamic envelope, and workspace hazard checking—adds a **mean latency overhead of only** 0.85 ms per execution cycle. This overhead comfortably fits within our allocated 3.33 ms System-1 real-time budget, confirming that deterministic functional safety can be enforced without sacrificing 300 Hz control stability.